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The rise and flows of blue carbon credits advance global climate and biodiversity goals



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Blue Carbon Projects conserve and restore degraded BCEs to mitigate climate change^{1–4}, primarily funded through carbon markets^{2,4–6}, thereby delivering climate finance^{6–8}. Our assessment of 81 BCPs globally (2.0 million ha) shows annual emission reductions of 20.4 MtCO₂e/yr, targeting 5.8 GtCO₂e by 2075—≈2% of estimated potential^{4,9}. Between 2014 and 2025, 6.96 million BCCs were issued and 3.65 million retired, mostly transferred from LDCs and developing to developed nations below real value.

Coastal blue carbon ecosystems (BCEs), including mangroves, tidal marshes, and seagrasses, play a substantial role in carbon sequestration, such that avoiding losses and restoring degraded habitats can help mitigate and adapt to climate change^{1–4}. Moreover, BCEs offer a wide range of ecosystem services that benefit the communities where they are implemented, including coastal protection, supporting biodiversity, enhancing food security, and contributing to sustainable development and climate adaptation on a global scale^{4,10}. Although BCEs cover <2% of the ocean area¹¹, BCEs store over 30,000 TgC (≈110.1 GtCO₂e), comparable to the total carbon stored in terrestrial forests^{4,12}, and remove between 235 and 450 TgC yr^{−1}¹² (≈0.862–1.652 GtCO₂e yr^{−1}), demonstrating their remarkable efficiency in carbon sequestration, five times higher per hectare than terrestrial forests². Coastal BCEs are responsible for ~50% of the carbon burial in marine sediments^{11,12}. Yet, BCEs are threatened by anthropogenic activities, which have resulted in the loss of about half of their historic extent¹. Avoiding further losses and restoring degraded BCE habitats can therefore contribute to mitigate climate change^{5,10,13–17} and help meet the Kunming-Montreal Global Biodiversity Targets¹⁸.

Carbon markets have emerged as a practical tool for providing the necessary funding for conserving and restoring nature, including BCEs. They quantify and verify the carbon sequestration resulting from conservation and restoration actions, and translate them into tradable carbon credits. Blue carbon projects (BCPs) facilitate a direct mechanism to monetize the environmental value of the carbon sequestration service of BCEs and enable the actions required to realize their climate mitigation potential while preventing and reversing biodiversity loss^{6,19,20}. They, therefore, help reducing the financing gap required for immediate and impactful ocean action^{5,17,19–22}.

The process to develop carbon credits from BCPs involves rigorous standards for verifying blue carbon credits (BCCs) by international registries that ensure the integrity of emissions trading in the voluntary carbon market²³. Businesses purchase carbon credits, including BCCs, directly from project developers or trade them through the voluntary carbon market to fulfill their corporate social responsibility and contribute to the United Nations Sustainable Development Goals (SDGs)^{24,25}. The revenues derived from the sale of these credits provide, in turn, financial resources, enabling source countries to restore and protect their BCEs from further degradation^{22,25}. Thus, BCPs protect biodiversity, support climate mitigation and adaptation, and sustain the livelihoods of local communities^{6,21}.

Yet, the voluntary carbon market has been impacted by a series of cases of carbon credits that were issued for land forest emission avoidance, but no carbon mitigation could be subsequently verified^{26,27}. These failures weakened investors' trust, leading to plummeting of carbon prices in the voluntary carbon market²⁸ at a time when the urgency to mitigate climate change and biodiversity loss is rapidly rising²⁹. As none of the cases above involved BCPs, BCPs have the advantage of a premium compared to carbon credits from traditional large-scale REDD+ projects⁵. While a recent study provides an extensive synthesis of market-based BCPs³⁰, our study offers an independent assessment by incorporating more projects and conducting a more detailed examination of implementation dynamics. Furthermore, our study follows carbon credit transactions and financial flows, with a particular emphasis on the movement of credits between developing and developed nations.

Here, we present a holistic overview of BCPs based on a thorough assessment of existing projects for which fundamental data were available and use this to explore the extent and rate of implementation, along with the BCCs demand-supply dynamics and their associated pricing. Our results

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demonstrate that BCPs are already providing an important contribution to both mitigating greenhouse gas (GHG) emissions and advancing the delivery of the Kunming-Montreal Biodiversity goals, based on financial flows from the developed to developing countries.

Rise of blue carbon projects

By March 2025, a total of 94 BCPs were registered across various international registries, including American Carbon Registry (ACR), Climate Action Reserve (CAR), Plan Vivo, and Verra, also known as Verified Carbon Standard (VCS), in addition to three projects in Japan, and two BCPs under Australian Carbon Credit Units (ACCUs) scheme which are registered in corresponding national registries. Amongst them, 19 projects have issued BCCs that are currently available in the market, with Verra being the most popular choice (8 out of 19 projects), although BCPs accounted for only 0.03% of Verra's credits by March 2025³¹. Thirteen projects have been canceled (rejected, discontinued, or canceled), with the remaining 62 projects pending approval [Supplementary Fig. 1]. From 94 BCPs, 67 (14 active and 53 inactive) projects have applied for VCS by Verra, with Plan Vivo being the second most common provider (four active and eight inactive). Lastly, ACR and CAR include six and two BCPs, respectively, of which one and two are active [Supplementary Fig. 2]. With Verra, 23 of the total and six of the active projects have also applied for another certification known as the Climate, Community & Biodiversity (CCB) standards. This effort is directed not only towards fulfilling climate-related ambitions but also to recognizing the co-benefits of projects in supporting local communities and contributing to biodiversity conservation³². A national approach has been developed in Japan, where BCPs align directly with governmental verification protocols, without involving international registry mechanisms, as their credits, J-bonds, are traded only in Japan³³. The Blue J-bonds include, uniquely, carbon credits against action involving seaweed restoration and conservation. Similarly, Australia has developed a distinctive regulatory framework for BCPs under the ACCUs scheme that integrates these projects within a formal national accounting framework. It ensures alignment with national climate policies and emission reduction targets while having rigidly defined methodologies for carbon sequestration assessment. The framework establishes scientifically robust methods for quantifying climate mitigation benefits and provides a streamlined process for project development and verification^{34,35}.

We excluded 13 canceled projects from further analyses to focus on the impact of current and planned BCPs. Most BCPs have a moderate scale of $26,019 \pm 63,351$ ha, with median size of 3000 ha [Supplementary Fig. 3]. However, there are a few BCPs with significantly larger sizes. In particular, Pakistan's "Delta Blue Carbon-1" project stands out as the largest BCP to date, covering 350,000 ha, 13.5 times larger than the average size of the remaining projects.

The size distribution of projects has varied over time. In 2011 and 2014, size distributions were quite similar, with median project sizes of 75,000 and 80,000 ha, respectively, and no new projects in 2012 and 2013. In 2015, a wider range of sizes was observed. Furthermore, the data reveal a significant increase in the sizes of BCPs in 2023, distinct from the previous 6 years that predominantly featured projects of nearly-demonstration scale, characterized by very small areas (Fig. 1a). BCPs have grown to encompass an extensive network of habitats, including 1.93 Mha of mangroves (72 of 81 BCPs), followed by 0.07 Mha and 0.007 Mha of saltmarshes (5 projects) and seagrass meadows (4 projects), respectively. The outstanding contribution of mangrove projects is even greater for estimated carbon sequestration, accounting for 99% of annual estimated GHG reductions from BCPs, resulting in an impressive estimated removal of 20.2 MtCO₂e per year by 2025 [Supplementary Fig. 4 and Supplementary Table 1]. Out of the 81 BCPs, only one did not provide information on the actions supporting additionality. Four (5%) of projects involved conservation of threatened BCEs to avoid emissions, while 30 BCPs applied both conservation and restoration, and the remaining 46 BCPs focused on restoration. The total area managed for conservation to avoid emissions was 309,372 ha (15.4%), while 842,293 ha (42.0%) were managed by a combination of conservation

and restoration, and 851,779 ha (42.6%) were managed through restoration alone. Notably, some voluntary carbon markets, such as the ACCU scheme, do not support blue carbon conservation projects and only recognize restoration efforts^{34,35}.

Because there is generally a lag between the starting year of a project and the year the project is verified or the year it enters the carbon market, the issuance year of credits is typically different from vintage year. The first BCP closely followed the presentation and widespread endorsement of the Blue Carbon concept at COP15 of the UNFCCC in Copenhagen, particularly by Indian Ocean nations¹². Its momentum further accelerated with the adoption of the Paris Agreement in 2015, which reinforced international recognition of nature-based climate solutions. However, the most significant surge in the delivery of BCPs occurred in 2021 and 2022, with 11 and 14 new projects, respectively (Fig. 1b).

Existing BCPs encompass 2,003,443 ha and an estimated committed reduction in GHG emissions by ~ 20.4 Mt CO₂e yr⁻¹ to be reached by 2026 [Supplementary Table 2]. The committed GHG emission reduction from existing BCPs by 2026 represents 2% of the estimated full potential of BCPs, but 2.8% of the full potential for the case of mangroves⁴, suggesting that the scope for growth of BCPs is 50 times their current scale. Based on estimates, there are ~ 36 –185 Mha of BCEs globally⁴; hence, a small proportion is being conserved and restored by BCPs. However, the contribution of BCPs to mangrove restoration is substantial, with the commitment to conserve and restore 1.93 Mha of mangroves representing an expansion of mangrove area by 14% of the estimated present area of 13.76 Mha³⁶. Hence, BCPs are already making a significant contribution to bend the curve of loss for mangroves and meet the global restoration targets for mangroves³⁷. The contribution of BCPs in reversing the loss of saltmarshes and seagrass meadows is modest, particularly so for the latter⁴. This highlights the potential for expanding conservation efforts and the importance of investing in BCPs to both advance GHG emission reductions and the restoration goals under the Kunming-Montreal Global Biodiversity Framework target of restoring 30% of degraded habitats by 2030³⁸. Failure to protect these valuable habitats not only diminishes their capacity to capture carbon but also increases the risk of their degradation and the emission of significant amounts of carbon^{4,15}.

Most BCPs are structured around durations of 20, 30, and 40 years, yet 30-year projects dominate in terms of both their spatial extent and GHG reduction capabilities (Fig. 1c), which might suggest a strategic balance between ecological impact and project manageability [Supplementary Table 3]. However, the median duration of BCPs—30 years—covers only 30% of the 100-year permanence³⁹, typically required for carbon removal to be considered fully effective in terms of climate benefits. This shorter duration may have contributed to the lower valuation of the carbon credits issued by BCPs.

The geographical distribution of BCPs' shows the dominance of projects in Asia and the Americas, primarily rooted in tropical countries that have abundant blue carbon habitats (Fig. 2a). Remarkably, Australia, one of the nations with the largest blue carbon resources⁴⁰, and Japan, both with national projects remains ineligible for international carbon standards, with projects able to generate credits only within national frameworks. The source countries span a range of socio-economic contexts, though the majority are classified as developing or least developed countries (LDCs) according to the UN. Mexico hosts the highest number of BCPs (11), followed by India (8) and Myanmar (7), but Pakistan, Nigeria, and Colombia, collectively represent more than half of the total area of BCPs, with individual contributions of 30%, 14%, and 12%, respectively of the total area by March 2025. Whereas Mexico has the highest annual average estimated GHG reductions (5.7 MtCO₂e yr⁻¹), followed by Pakistan (4.4 MtCO₂e yr⁻¹) and Indonesia (2.7 MtCO₂e yr⁻¹) from BCPs they support [Supplementary Table 4], the large mangrove project in Pakistan surpasses the others in accumulated emission reductions over its 60-year project lifespan (Fig. 2b). The BCPs provide an estimated cumulative removal of ~ 5.8 GtCO₂e by 2075 (Fig. 2c), highlighting their potential contribution to climate change mitigation.

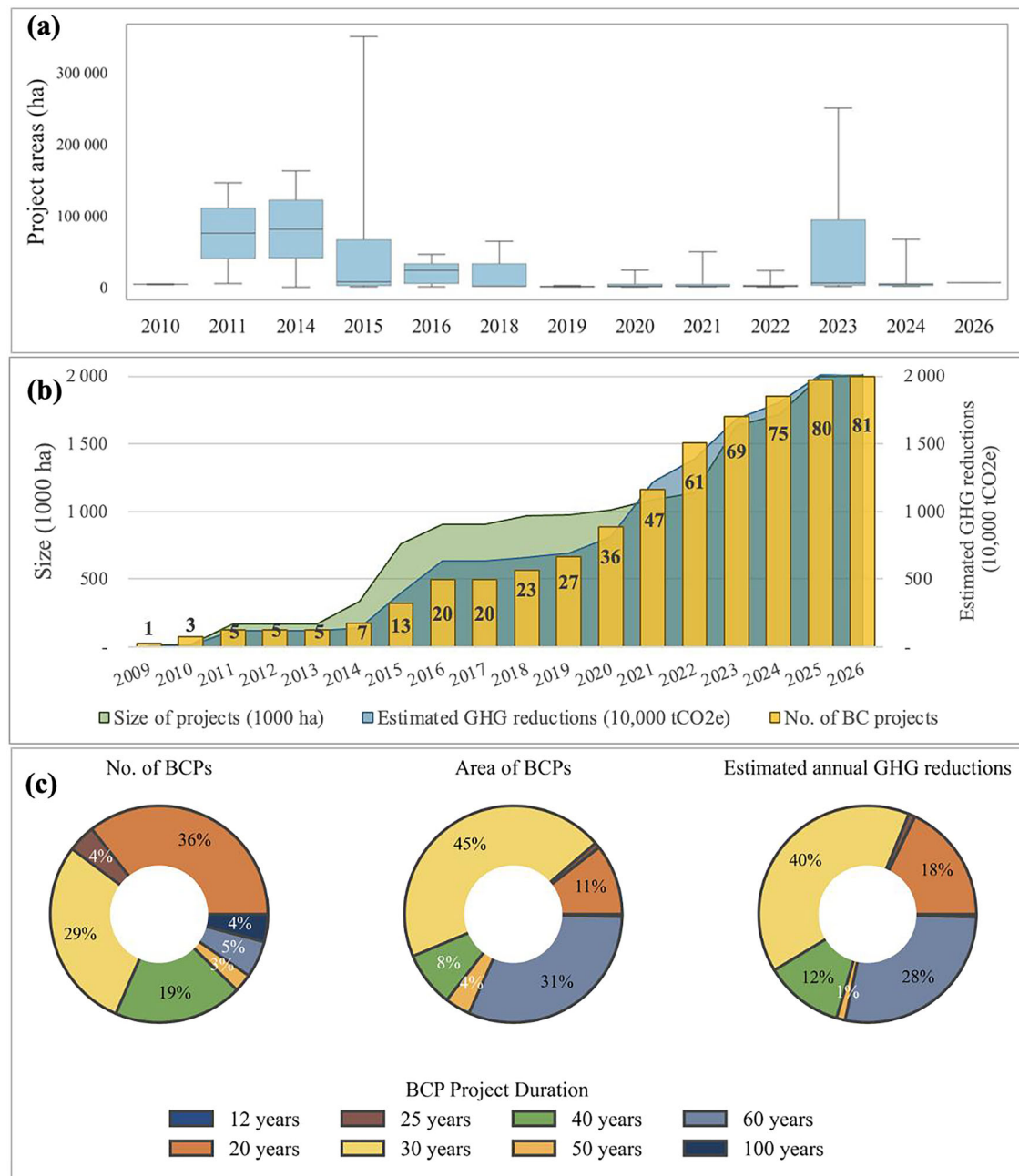


Fig. 1 | Comprehensive overview of BCPs dynamics. **a** Box plots of BCP area distribution by project start year (2010–2026), with the central line indicating the median, boxes representing the central 50% quantile, and whiskers covering the full data range. **b** Growth of BCPs (2009–2026). Stacked green and blue plots show

accumulated area (left axis) and estimated annual GHG reductions (right axis), with project counts indicated by yellow bars, based on documented start years. **c** BCP lifespan analysis. The pie charts illustrate distribution of committed duration by project count, area, and annual GHG reductions.

The flow and value of Blue Carbon Credits

Excluding Japan and Australia, only 12 among the 25 countries identified as sources of BCPs, had issued BCCs by March 2025 (Fig. 3a, b). BCCs contributed only 0.35% and 0.91% to the total and to nature-based solutions (NBS) carbon credits issued in the market, respectively^{22,41}. The NBS category includes three main sectors: agriculture, forestry, and wetlands, with BCPs falling under the wetlands subgroup. In 2014, a total of 159,493 BCCs were issued, marking their introduction to the carbon market. The issuance trend has shown a steady upward trend, with the number of issued credits surpassing 1 million by 2020 and then continuing to rise sharply following the entry of Pakistan's project into the market in 2022, reaching about 6.96 million credits by March 2025 [Supplementary Fig. 5, Supplementary Table 5]. Pakistan holds a 69% share in the BCC market, and its combined

market share with India (8%), Senegal (7%), and Indonesia (6%) amounts to ~90% of BCCs issued to date [Supplementary Fig. 6]. All countries issuing BCCs are classified as developing nations or LDCs, except for the US (BCCs from Japan and Australia are not included here due to the lack of available data). According to the most recent data, the average Human Development Index (HDI) of these countries is 0.645, which is below the global average (0.756). China records the highest HDI of BCC source nations at 0.797, ranking 78th globally, whereas Madagascar has the lowest at 0.487, positioned at 183th out of 193. Significantly, four of these countries are in the low human development category⁴². Consequently, revenue from BCCs plays a critical role in financing both the conservation and restoration of their BCEs, as well as enhancing local community livelihoods^{24,43}.

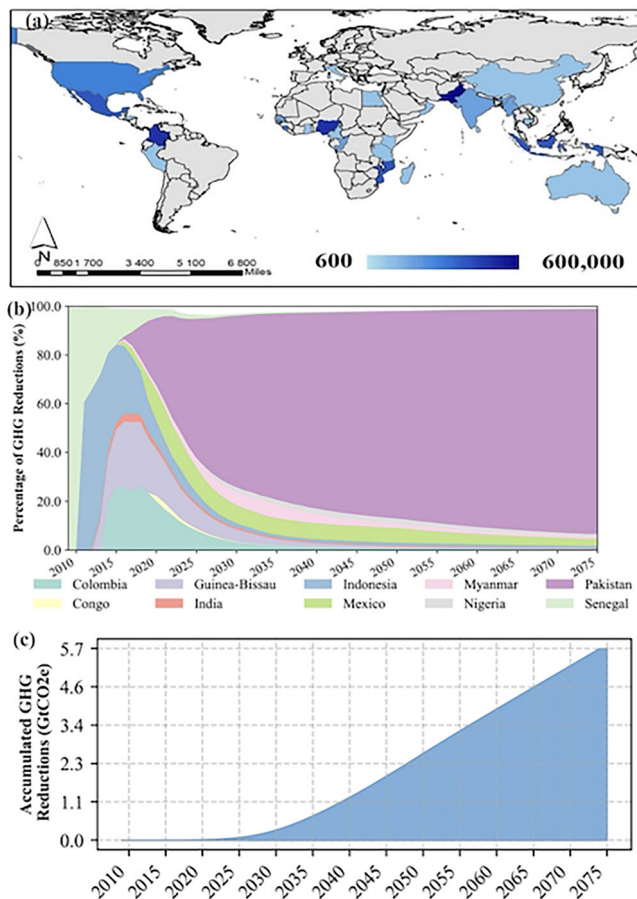


Fig. 2 | Global distribution and contributions of BCPs. **a** Global distribution of BCPs. The map indicates project sizes across countries, with darker colors denoting larger projects. **b** Country-specific GHG reduction contributions. The plot illustrates the cumulative share percentage of each country per year, represented as an area graph to better visualize the contribution trends over time. **c** Total estimated GHG reductions (GtCO₂e) of BCPs. For each year, we have aggregated GHG reductions of projects within their respective timelines, as provided in projects documents.

When analyzing BCCs by their vintage year (i.e., the year in which the underlying carbon sequestration occurred), we observe an exponential increase until 2021, followed by a significant drop in 2022. Only a few credits were linked to vintages in 2023, and none appear for subsequent years. By 15 March 2025, a total of 6,961,950 BCCs were introduced into the market, with 3,651,383 of these credits retired and removed from circulation (Fig. 4a), indicating that ~48% of the issued BCCs are still in circulation. Due to the COVID-19 pandemic, BCC retirements temporarily declined between 2019 and 2021, but an upward trend has since resumed (Fig. 4b). In terms of issuance year, BCCs first entered the carbon market in 2014. The number of BCCs decreased by half in 2015 but then showed a consistent upward trend in the following years until 2019, when it experienced a sharp decline. However, BCC issuance increased significantly in 2020, continuing to rise in 2021, and reached its peak in 2022, driven by the issuance of BCC from Pakistan's large-scale mangrove project. The number of credits issued in 2023 was comparable to 2021 levels, with a significant rise again in 2024 (Fig. 4c).

BCCs find buyers among corporations, individuals, governments, non-governmental organizations (NGOs), and investors worldwide, to offset emissions and demonstrate their commitment to mitigation goals⁷, by retiring BCCs⁴⁴. Out of the 1,109,210 BCCs that we tracked, more than 97% were purchased by entities from developed and high-income countries such as the US (42%), European countries (40% with a higher concentration of

purchases in the UK and France), Japan (8%), and Canada (7%) (Fig. 5, Supplementary Fig. 7 and Supplementary Table 6). Countries demanding BCC have an average HDI of 0.917, significantly higher than the mean HDI of source countries (0.645)⁴².

Market demand for BCCs is increasing exponentially [Supplementary Fig. 8]. The primary buyers are large-scale entities in various sectors such as technology and software giants (like Apple and Microsoft), transportation (shipping and airlines), pharmaceutical companies, financial and insurance sectors, as well as the steel industry. Notably, some BCPs do not openly trade credits; instead, credits are used to offset the financing company's carbon footprint. "Mangrove Restoration and Coastal Greenbelt Protection in the East Coast of Aceh and North Sumatra Province", Indonesia is one such example¹⁷. Importantly, not all BCCs are purchased directly by the end buyers; a considerable portion (about 20%) is bought by resellers for speculation, who subsequently resell these credits at higher prices than those allocated to the BCPs' developers. Unfortunately, resellers do not disclose the final destination of credits, obscuring the final destination and value of 20% of the credits in this study, which is based on the dataset we compiled. Reseller organizations trading BCCs are mostly located in the US, UK, France, Sweden, Switzerland, and Finland (Fig. 5). For instance, ~78% of the BCCs flowing to the UK have been purchased by resellers, whose final destination remains unknown. This share for resellers is considerable and can influence the market price, potentially leading to disproportionate revenues for resellers relative to communities delivering the projects. The development of a fair voluntary carbon market requires full transparency in credit pricing—both in payments made by primary buyers to project owners and by final buyers to intermediaries in resale transactions. A mandatory register of projects under the UN Carbon Offset Platform⁴⁵, along with integrated digital infrastructures—such as the International Carbon Registry⁴⁶—can help enhance transparency in the voluntary carbon market.

BCCs mostly flow to the Global North, where cumulative GHG emissions have been greatest, from the Global South, where local communities strongly depend on ecosystem services from blue carbon habitats, while also relying on extractive activities, such as building and agriculture or aquaculture, to derive value from these habitats in the absence of BCPs. Among source countries, Pakistan, Myanmar, and Kenya have sold their BCCs to a wider range of countries, while the BCCs buyers of Senegal, India, and Madagascar were more geographically restricted. Credits originating from projects within China, Australia, and Japan have been sold exclusively within their national borders (Fig. 6).

The dynamic interplay between the suppliers and purchasers of BCCs fundamentally influences BCC pricing. Carbon markets are broadly categorized into two types: compliance carbon markets (CCMs), which function under regulatory frameworks mandating emission reductions, and voluntary carbon markets (VCMs), where actors engage in carbon offsetting at their own discretion, without legal compulsion⁴⁷. BCCs are mainly exchanged in the VCM through Over-The-Counter (OTC) transactions, which are direct, private deals between parties based on verified credits. These deals are formalized through Emission Reduction Purchase Agreements (ERPAs), which outline the terms of the trade, such as quantity, pricing, and delivery of credits, along with any financial arrangements. ERPAs record the agreed price per credit for VCM transactions²⁹. Negotiating the prices of BCCs involves an economic mechanism between the buyer and seller. Several factors, including ecological value, carbon sequestration potential, and market liquidity, affect the market value of BCCs. The bargaining power of the involved parties is an important factor in shaping VCM prices. It is not uncommon for buyers or groups with a large market share to exert significant influence over pricing, highlighting the role of power imbalances in setting the prices within the VCM^{22,29}.

Each BCC corresponds to one metric ton of CO₂ equivalent (t CO₂e). Published reports indicate BCCs are valued between USD 5 and 35 per tCO₂e, with mangrove projects commanding higher prices^{22,29,48}. There is currently no accurate or reliable price information available on BCC pricing for many BCPs, and we were able to obtain data for only five projects registered by Plan Vivo and Verra, in addition to the project in Australia and

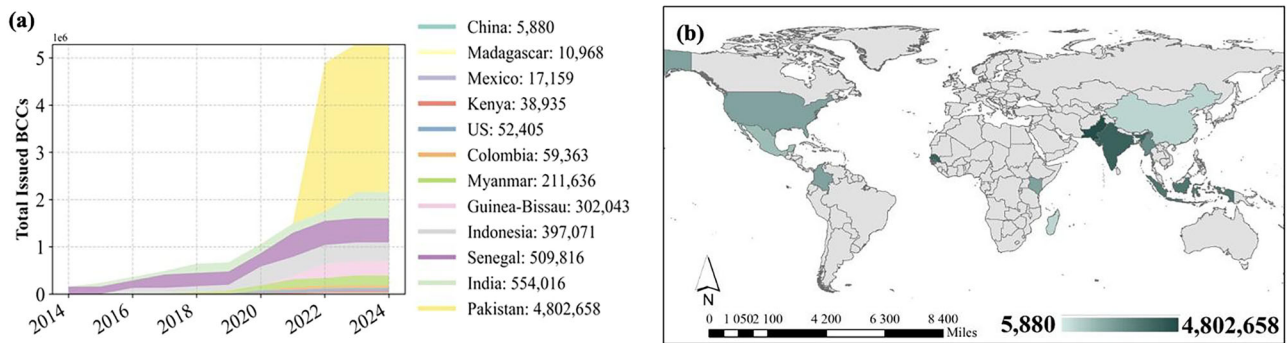


Fig. 3 | Supply dynamics of BCCs: Issuance trends and geographic distribution (2014–15 March, 2025). **a** Accumulated BCC issuance trends. The graph shows accumulated BCCs issued by country spanning from 2014 to March 15, 2025 identifying key contributors. Data are based on the year credits were issued, not the

vintage year. **b** Geographic Distribution of issued BCCs. The color-coded map indicates total issued BCC by March 15, 2025, with darker shades representing more BCCs issued.

three BCPs in Japan. For Plan Vivo projects, the price started at around USD 6 in 2014 for a project in Kenya and has gradually increased, reaching approximately USD 19 in 2023. Project-specific average prices ranged from USD 12.3 to USD 18 per BCC^{49–51}. In comparison, limited data from VCS projects in 2022 showed prices at about USD 27.8 per BCC for Pakistan⁵², and around USD 15 per BCC for a mangrove project in Colombia [Supplementary Table 7].

Price disparity between BCCs and carbon credit prices in compliance markets is substantial. For example, EU-ETS's price was USD 6.76 in 2014, surging to USD 86.5 in 2022 and further to USD 96.3 in 2023 and USD 61.3 in 2024⁵³ [Supplementary Fig. 9]. Amongst suppliers of BCCs, only three—the US, Japan, and Australia—are classified as developed. We were unable to obtain pricing information for BCCs for the U.S. market. In Japan, however, each BCC is exchanged at a price ranging from 8000 to over 13,158 yen (USD 72.8–119.8) within the country²⁵, which is significantly higher than carbon price under the Tokyo Cap-and-Trade Program (Tokyo CaT)—Japan's regional emissions trading scheme—where prices averaged USD 49.1 in 2023 and USD 36.9 in 2024⁵³. In Australia, BCCs are traded at approximately USD 22.1 per unit⁵⁴, approximately equal to the 2024 carbon price under the Australian Safeguard Mechanism (USD 21.9)⁵³, a regulatory framework that sets emissions baselines for large industrial facilities and allows compliance through the purchase of carbon offsets.

A weighted average price of USD 26.96 per BCC was calculated using available price data from individual projects, weighted by the number of BCCs issued. This value was then extrapolated to all projects. Assuming that existing and pending BCPs collectively sequester 60.9 Mt CO₂e by 2024, the estimated investment generated from BCC sales could reach approximately USD 1.62 billion. This provides a first order estimate of the size of the market, which is conservative as carbon credits reaching the buyers from resellers likely sell at a much higher price than the average price used to generate this estimate, an assumption that needs to be verified when increased transparency in the future allows to assess this hypothesized increase in price.

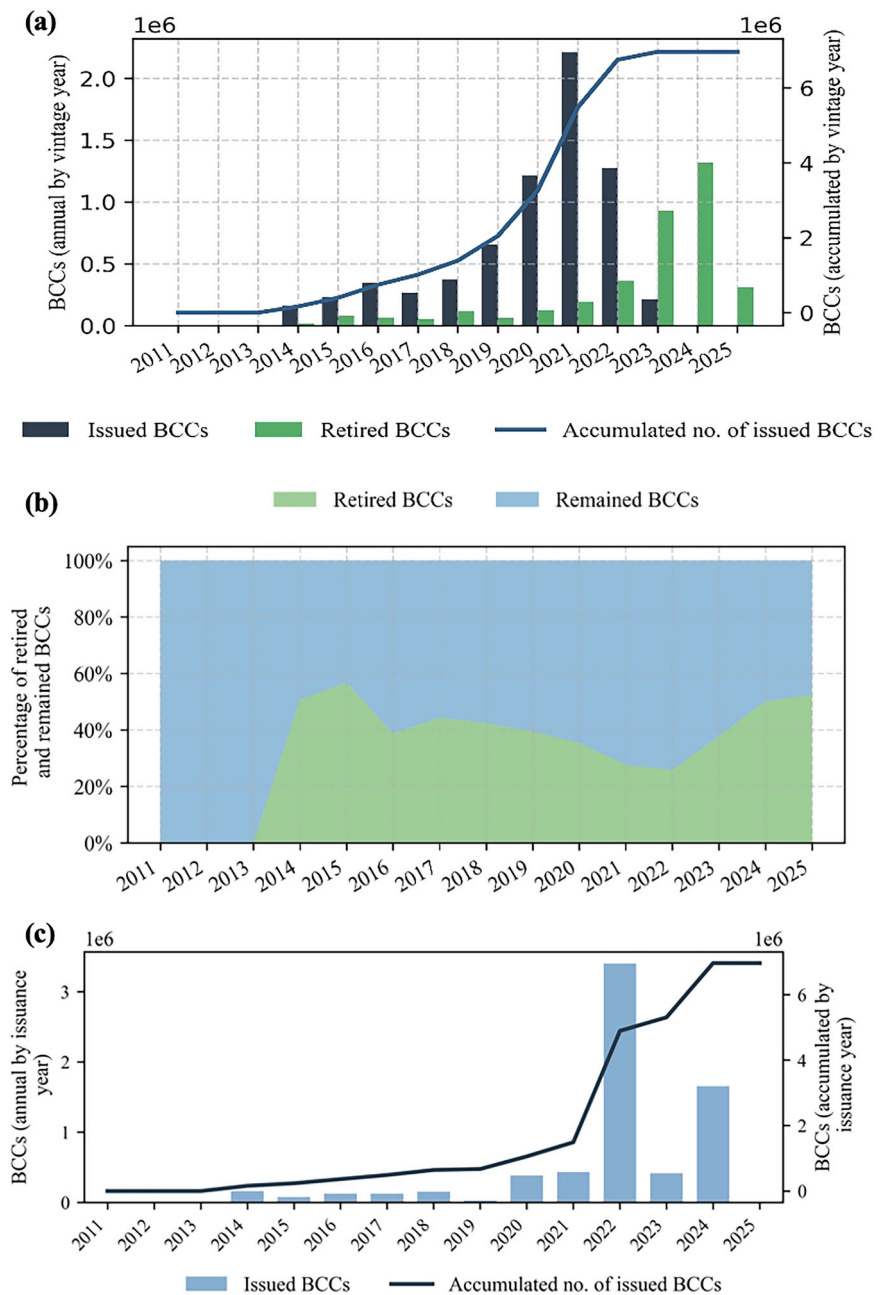
In most source countries, Indigenous Peoples and local communities actively participate in the projects, with revenues contributing to their livelihoods²⁹, particularly so when located in either LDCs or developing nations. In contrast, the purchasing entities predominantly reside in wealthy, developed countries. Therefore, exchanging BCCs at prices below their real value is unfair and detrimental to the Global South, where BC resources originate. This price disparity reflects broader economic and environmental inequalities, aligning with the climate debt theory, which attributes historical responsibility for environmental damage to high-emitting nations at the expense of low-emitting nation⁵⁵. Industrialized nations are estimated to have overused their portion of the earth's carbon budget, accumulating a climate debt of approximately US\$192 trillion to developing nations as reparation for over emissions⁵⁶. Underpriced BCCs further perpetuate this inequality, in effect allowing the Global North to use environmental assets of the Global South without just compensation.

This creates a major inconsistency, whereby buyers in Europe and the U.S. buy BCCs from Kenya and Madagascar at ~25% or less of the regulated value of carbon credits within those nations. In fact, the highest value for carbon credits, by far, was assigned to internally traded BCCs generated by BCPs in Japan. The weighted average price paid for BCC was approximately USD 26.96 for 2022–2023^{49–51}, compared to the estimated value of carbon in the buying countries of about USD 54.3 (2022–2023)⁵³, indicating that buyers are paying nearly half of what they would if acquiring credits locally. Accordingly, the initial step toward fostering a more equitable market involves gaining a thorough understanding of BCPs and constructing a holistic representation of the BCC market dynamics. As the EU imposes the Carbon Border Adjustment Mechanism (CBAM), taxing imported goods on the carbon impact, it should require carbon credits to be purchased at the point of origin at a level equivalent to taxation rates. For equity, countries like the U.S. and the UK should implement comparable policies, requiring imported carbon credits to be purchased at their recognized worth. Similar policies in compliance markets, such as California's Cap-and-Trade Program and Singapore's Carbon Tax, bring carbon consistency across jurisdictions. Adopting similar mechanisms in voluntary carbon markets could prevent BCCs undervaluation and promote fair pricing.

Whereas several assessments have estimated the potential flow of BCCs from BCPs, and a recent analysis³⁰ has analyzed existing projects, our study builds on and extends these efforts by incorporating an updated dataset and providing a detailed examination of real-market transactions, financial flows, and price disparities. Given the critical role of NBSs⁵⁷, our findings confirm the rapid development of BCPs. While BCPs are growing in significance, they still represent a relatively small portion of the total NBS-related credits. The committed restored area, along with annual and cumulative emission reduction, is substantial, particularly for mangroves. These efforts contribute meaningfully to both climate and biodiversity goals. Further expansion of BCPs is needed to meet their estimated potential of BCPs to offset between 0.5 and 1.38 GtCO₂e yr⁻¹ by 2050⁹, with an additional 0.84 (0.62–1.06) GtCO₂e yr⁻¹ expected by 2030 through restoration efforts⁴. Efforts must be made to accelerate the development of BCPs for saltmarshes and, particularly, seagrass meadows, where very large projects, which could potentially surpass the largest existing mangrove project, are being developed for the Bahamas⁵⁸. In addition, seagrass-based carbon projects are being developed in Japan, the US, Spain, France, Italy, and the UK, and elsewhere, supported by a number of emerging seagrass-specific carbon methodologies, such as Verra's VM0033 methodology for tidal wetland and seagrass restoration⁵⁹.

BCPs are efficient, robust, and viable tools for preserving and restoring BCEs, playing a crucial role in the global effort to combat climate change and delivering a broad range of co-benefits. Our analysis does not provide evidence that recent scandals affecting avoidance projects on forests to have impacted the development and price of carbon credits from BCPs^{26,27}. Indeed, many of the projects generate mitigation additionally through

Fig. 4 | Dynamics of BCCs supply (2011–15 March, 2025). **a** Issued and retired BCCs with accumulated trend by vintage year. Dual-axis chart depicts annual issued and retired (bar charts, left axis) against a trend line of accumulated issued BCCs by vintage year (line graph, right axis). **b** The proportions of retired and remained BCCs by vintage year. Stacked bar chart illustrates the distribution of retired versus remained BCCs, offering insights into market absorption. **c** Annual issued and accumulated BCCs by issuance year. The dual-axis chart depicts annual issued BCCs (bar charts, left axis) against a trend line of accumulated issued BCCs (right axis) by issuance year.



restoration efforts, which are far more robust and offer more reliable verification pathways compared to projects issuing credits based on avoidance of emissions resulting from threats to the conservation of BCEs, which all the forest projects involved in the scandals were based upon.

A recent assessment established that, as of the end of 2024, around 2.40 billion carbon credits have been issued in the voluntary carbon market, of which around 56% have been retired⁶⁰. These credits derive from an inventory of 11,302 carbon projects developed in 143 countries, with emerging and developing countries generating 75.13% of the total carbon credits⁶⁰. NBS projects generated 38% of the total inventory of credits issued in the voluntary market⁴¹. BCPs contributed 0.91% of the NBS credits issued in the voluntary carbon market. While their later entry, beginning in 2014, may partly explain this modest share, other contributing factors likely include methodological complexity, limited eligible ecosystems, and the slower development of standardized protocols, all of which have constrained their growth relative to land-based NBS projects that have been active since 1996.

We showed that, if all existing and pending blue carbon projects fully issue their expected carbon credits, investments in BCCs could amount to at least USD 1.62 billion, highlighting their growing role in climate change mitigation efforts. BCCs are increasingly recognized for their potential to contribute to biodiversity conservation, shoreline protection, food security, and community well-being⁴. The Kunming-Montreal Global Biodiversity Framework encourages nations to activate NBSs, such as BCPs, to the maximum ambition possible, recognizing the importance of the co-benefits they deliver. Though outstanding efforts have been made regarding BCPs in order to conserve and restore BCEs, projects need to grow 50-fold to meet their full potential, thereby releasing resources to protect BCEs⁶¹. BCPs, as a mechanism within the voluntary carbon market, represent a useful and promising path to channel financial flows from developed to developing nations to help meet their biodiversity goals for BCEs. However, these projects should generate equitable and fair financial flows. Yet, our analysis reveals a consistent asymmetry between the value of carbon credits from BCEs sourced from LDCs and developing nations and the value of carbon

Fig. 5 | Global distribution and dynamics of BCCs demand (2014–15 March, 2025). The color-coded map indicating the volume of purchased BCCs (darker = higher volume), and pie charts show the share of end-buyers versus resellers.

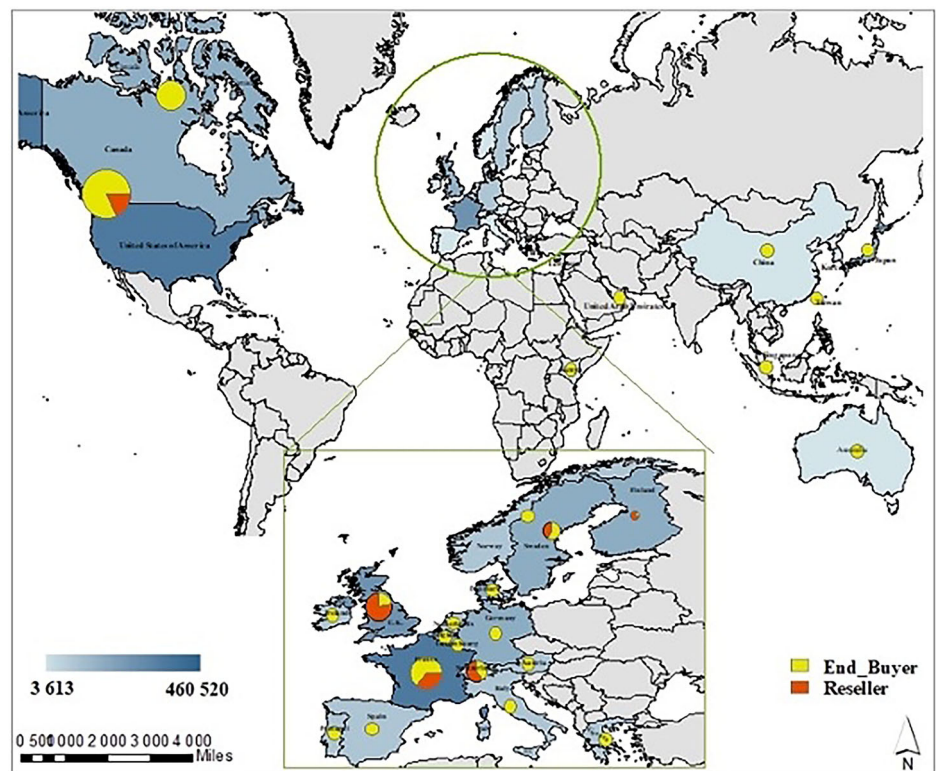
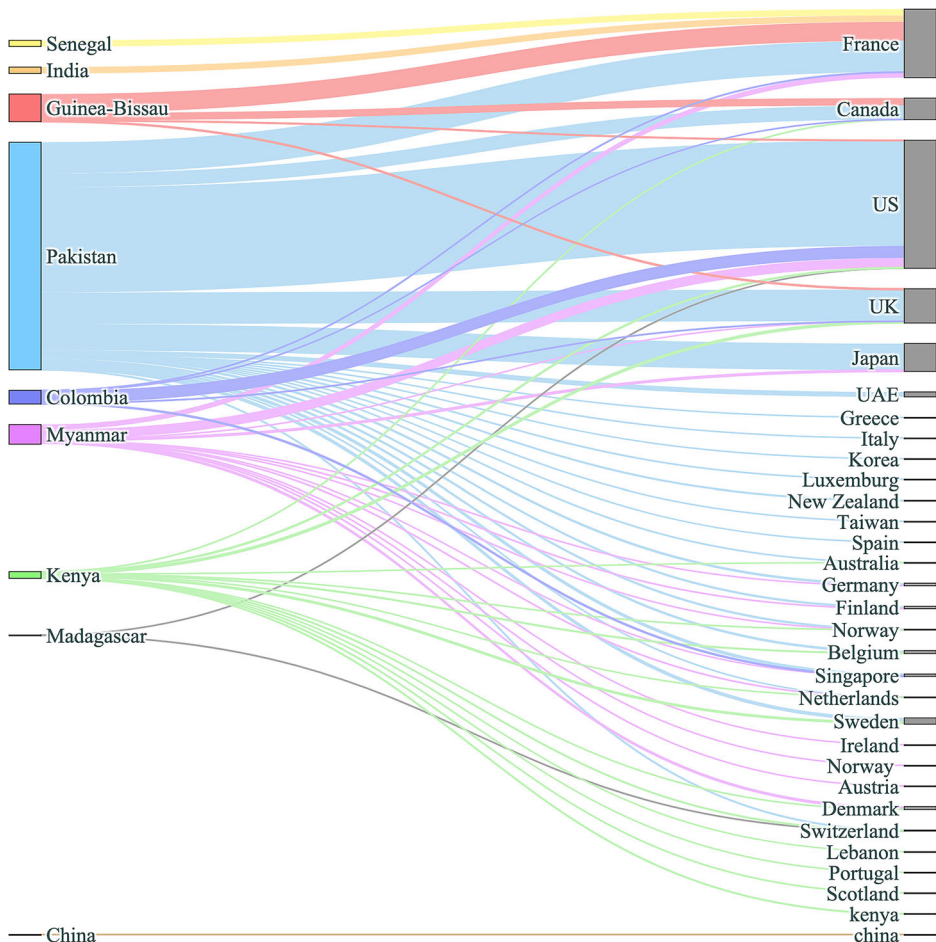


Fig. 6 | The global flows of BCCs trading. The links connect source and buying nation, with varying link thickness scaled to the carbon credit volume traded. Australia and Japan have been excluded from this diagram due to the lack of data.



regulated in receiving developed nations. This pattern reflects more profound inequities in the VCM, where BCCs are often purchased below levels observed in compliance systems. To ensure equity and support SDGs, BCC markets require more transparent and regulated pricing frameworks, not only to ensure price parity between developing and developed nations, but also to recognize the true value of BCPs to advance SDGs. Projects should, in the future, recognize, document, and quantify the full spectrum of benefits derived from BCPs, which include both ecological and social benefits addressing multiple SDGs, and build a premium on the price for the contributions of the BCPs to rebuild natural and social capital.

We appeal to regulators of nations importing carbon credits to demand transparency in reporting by the projects, including establishing transparent registers to track final buyers beyond resellers, reducing speculative trading, and ensuring the disclosure of prices paid by final recipients for BCCs. Increasing BCPs transparency is key to supporting and building BCPs under strengthened governance that will ultimately harmonize voluntary and compliance carbon markets. Enhancing the transparency of BCPs and avoiding the current gap in carbon price paid to developing nations by importers relative to that in their markets will contribute to stabilizing the emerging yet crucial blue carbon market while enhancing investor confidence both in the quality of the projects and their ethical integrity.

Furthermore, an essential consideration is that BCC pricing must adequately cover the costs associated with implementing and maintaining BCPs to ensure their long-term viability. Recent research highlights the differences of carbon pricing, and the cost-effectiveness of BCPs, and shows the gap between the price of credits on the voluntary market and the true worth of the credits⁶². Research in the future should continue investigating mechanisms for realizing equitable pricing frameworks that capture both the carbon sequestration potential and extensive ecological and socio-economic advantages of BCPs.

Methods

Data collection

As a crucial part of our study, we spent considerable time collecting information about BCPs and BCCs. The outcomes of this phase are expected to be valuable and will undoubtedly play a significant role in the success of our project. However, this task was quite challenging due to the limited data availability, particularly regarding BCC prices and potential buyers.

Data collection for BCPs

The list of BC projects was extracted from different registries. Each registry maintains its own standards, formats, and accessibility, requiring us to develop customized data extraction and normalization approaches. We engaged with multiple registries globally, each hosting distinct BCPs. For the ACR registry, in the projects list with 727 registered projects, there were six projects of the type “Wetland Restoration”⁶³, of which only one project was active. For the CAR registry, from 911 registered projects⁶⁴, we found two BC projects by searching different types of wetlands in the list of projects because there was no separate category for wetlands in this registry. For Plan Vivo, we checked all 28 current projects⁶⁵ and found four active BC projects^{49–51}. We also found 10 projects in the list of pipeline projects⁶⁶ in this registry by searching related keywords. BCPs registered by Verra (VCS) are grouped under Agriculture Forestry, and Other Land Use (AFOLU). Therefore, we searched by wetland types and verified methodologies for wetlands to ensure we had retrieved all relevant projects⁶⁷. According to our final search results, we found 67 BCPs registered by Verra.

After listing BCPs, we extracted data from published reports. We went through each project’s registry-provided documents individually. All data relating to the projects, including the type of wetland, the size, location, the average estimated annual GHG reductions, the start year of the project, the duration of the project, and the estimated reductions by year throughout the project’s lifetime, were retrieved from each project’s documents.

In total, 94 BCPs were listed by different registries. Of those, 13 BCPs were canceled. Among these, three BCPs were canceled within ACR projects, and two were discontinued under Plan Vivo. The remaining canceled projects

were under Verra, with reasons for cancellation including inactivity, withdrawal, denial of registration requests, rejection of registration and verification approvals, and administrative rejection. Therefore, this paper focused on the 81 projects that are currently active or under review and verification.

Data collection for BCCs

For BCPs registered by ACR and CAR, the data on BCCs were extracted from the international dataset of Berkeley Voluntary Registry Offsets Database⁶⁸. Annual project reports provide data related to the BC market for projects registered by Plan Vivo. We manually checked all annual reports for three active projects to collect data on BCC’s issuance and sales. The buyer lists were then compiled from these reports^{49–51}. For Verra’s projects, the data on BCC issuance and retirement are provided by Verra as Excel files, which are downloadable for every project.

Once we finished compiling the final list of over 200 buyers who purchased BCCs, we conducted a comprehensive online search to gather additional information about them. This search involved identifying their location, type of activity, and the purpose behind their purchase of carbon credits. The aim was to collect more relevant data about the buyers and to enhance our understanding of their profiles. This step proved to be challenging. However, it is important to note that we needed access to the complete list of BCC purchasers, and the only available data are for three and two projects registered by Plan Vivo and all projects registered VCS, respectively. In the context of Plan Vivo projects, from 49,903 issued credits, 30,917 have been sold, and data on purchasers is accessible for 94% of these traded BCCs, in contrast to 53% for Verra-related projects (VCS has issued 6,842,483 BCCs of which 3,610,098 have been retired). Consequently, the analysis is based on the obtainable data concerning the buyers of 1,109,210 BCCs. Accordingly, due to the limited data available, the analysis of the demand side of the BC market might be constrained.

For the price of BCCs, we encountered difficulties due to limited data availability. However, we managed to estimate the BCCs’ price of Plan Vivo projects by analyzing data on their annual revenues and sold BCCs. Also, very limited data was found for only two Verra projects. No price data is available for ACR and CAR projects. For the Japanese projects, we had the price information in ¥²⁵, and we converted it to USD according to the conversion rate of ¥/\$ in 2021 (0.0091)⁶⁹. For Australia, price information was available in AUD⁵⁴ and was converted to USD using the 2024 average exchange rate of 0.6596 AU\$/US\$⁶⁹, corresponding to the last year in which BCC transactions were reported. Consequently, another limitation of the study was the lack of available price data for BCCs. If this data becomes accessible, a more precise analysis of the market can be conducted.

Data analysis, visualization, and mapping

In constructing a comprehensive dataset for analyzing the blue carbon market, we integrated all data retrieved from project reports and registries into a spreadsheet, which included information like the type of wetland, registry, source country, voluntary status, methodology (protocols), start year, and duration for BCPs and country, registry, vintage year, and issuance year for BCCs.

Given the formation of new BCPs and the dynamic nature of the BCC market, our dataset was systematically updated to include the latest information and projects up to 15 March 2025. This continuous updating process not only enhanced the robustness of our findings but also reflected the real-time fluidity of the BC market.

Data visualization and analysis were conducted in Python 3.9⁷⁰, using the Jupyter Notebook environment⁷¹ via Anaconda, and supported by standard libraries including Pandas⁷², Seaborn⁷³, Matplotlib⁷⁴, and Plotly⁷⁵.

Geographical distributions for BCPs and BCCs supply and demand were mapped using a global basemap⁷⁶ and ArcGIS, ArcMap version 10.8⁷⁷.

Data integrity and accuracy

Following data extraction, we integrated and standardized the information from various sources into a unified database. We cross-validated the data accuracy by systematically comparing information across independent

sources, including registries and BCP documentations, to confirm consistency and accuracy. Manual verification involved spot-checks and detailed reviews of a random sample of the data entries, comparing them against sources for accuracy. We carefully verified all data entries and calculations to ensure the accuracy and reliability of the data. Regular audits were performed to check data consistency and correctness across the dataset. Such rigorous verification processes safeguarded the integrity of the analysis, underpinning the study's findings with robust data.

Data availability

The data supporting the findings of this study will be made publicly available in an open-access repository upon publication.

Code availability

No specific code for results is provided for this study, other than the Sankey diagram, which will be available on request from the corresponding author.

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Author contributions

C.M.D. and N.H. secured the funding. C.M.D. and N.H. conceived the research. S.F., N.H., and C.M.D. designed the research. S.F. produced the data. S.F., N.H., and C.M.D. analyzed the data. S.F. wrote the first draft. S.F., N.H., and C.M.D. improved the text and approved the submission.

Competing interests

The authors declare no competing interests.

Additional information

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